Adjoint and shape-gradient validation in JAX-DSMC

Every gate, study and finding, 2026-09-05 to 2026-10-11

2026-10-11

How to read this deck

Arbiters, from strongest to weakest:

- analytic or exact references (closed forms, Rankine–Hugoniot, exact quadrature, replica-exact value gradients on collisionless cases);
- **CRN FD**: central finite difference on the *same* seeds (common random numbers), paired with the estimator seed by seed; noise grows like $1/h$ across hit-set jumps;
- **FD-only**: finite differences over many independent seeds (no pairing).

z = (estimator − arbiter) / SE of the paired difference. "Seeds" always means independent forward runs. Lesson learned repeatedly: never conclude from 2 seeds.

Timeline

dates	phase	part
09-05 – 09-11	forward gates; trajectory (DiCE) adjoint, boundary adjoints	I, II
09-14 – 09-17	estimator benchmark; P3 ensemble adjoint, route (b), virtual partners	II
09-18 – 09-24	embedded bodies; shape adjoint; level set; diffuse terms; cluster studies	III
09-25 – 10-02	`mode2` and squash studies on the cluster	III
10-02 – 10-04	bias study: seven findings, collisional terms built	IV

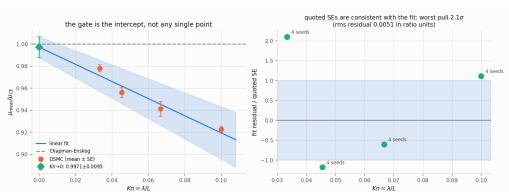
Derivation of the current estimator: `shape_derivation.tex` (with code: `shape_derivation_code.tex`).

Forward gates: transport and shocks

Shear viscosity, Couette (Chapman–Enskog, no free parameter). Four wrong explanations of an apparent deficit preceded the right one; every one was a quantity read without an error bar. The gate is the double extrapolation, 4 seeds per point:

$$\mathrm{Kn} \to 0: \ 0.9971 \pm 0.0095, \qquad dx \to 0: \ +0.0135 \pm 0.0039, \qquad \mu/\mu_{\mathrm{CE}} = 1.0105 \pm 0.0103 \ (|t| = 1.0).$$

2-D Taylor–Green decay: viscosity $+0.38\,\%$ from Chapman–Enskog ($R^2 = 0.9994$).
Normal shock (Bird §12.11): $L/\lambda_1 = 5.98/6.12/6.18$ at $D = 1/2/3$ against 6.2; jump conditions 0.1–$0.7\,\%$, shock speed $0.3\,\%$.
Inflow-driven Mach-2 shock: all six moments within $0.5\,\%$ of Rankine–Hugoniot, mass flux to $0.06\,\%$.



Forward gates: embedded bodies against SPARTA (2026-09-18)

Cut-cell geometry identical to SPARTA's (cell classes, cut set, volumes to 10^{-7}); argon circle and sphere, 4 seeds each, every tracking diagnostic zero.

gate	circle	sphere	criterion
final particle count	$z = -1.05$	$z = 0.97$	$\lvert z\rvert \le 2$
total drag	$+0.31\,\%$ ($z = 1.36$)	$-0.11\,\%$ ($z = -0.30$)	$\lvert z\rvert \le 2$
per-cell n, u, T	rms z 1.2, 90–92 % within 2	rms z 1.24, 91 %	
polygon vs analytic body	drag $+0.5\,\%$	$+0.75\,\%$	

Free-molecular drag (FG5), extrapolated to zero blockage, against the half-range surface integral:

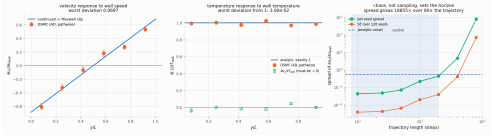
case	measured C_D	reference	z
circle, specular	17.080 ± 0.080	17.175	-1.18
circle, diffuse	20.429 ± 0.087	20.429	0.00
sphere, specular	14.684 ± 0.211	14.804	-0.57
sphere, diffuse	20.990 ± 0.295	21.211	-0.75

The reference box (60 % blockage) raises the drag $\sim$50 % above the isolated body's.

The trajectory (DiCE) adjoint, M6–M8 (2026-09-05 . . . 09-11)

Score-function plus pathwise AD through the whole run (`jax_dsmc/adjoint.py`).

- Oracles: [P1] Algorithm 2 and [P2] Algorithm 3 against `jax.grad` exactly (10^{-11}, 10^{-10}); move→collide recursion to 10^{-16} (Oracle 3).
- Pathwise wall gradients: $\mathrm{d}J/\mathrm{d}T_{\mathrm{wall}}$, $\mathrm{d}J/\mathrm{d}u_{\mathrm{wall}}$ to 3×10^{-12} against fixed-key FD, and against the Couette closed form (figure).
- Inflow `nrho` (score): FD/AD $= 1.04$, $|z| = 0.81$.
- **Attribution bias**: pairing each score with its own particle's reward is exact without collisions ($|z| = 0.18$) and $2.8\times$ low with them ($|z| = 22$), growing with the horizon (FD/AD $1.13 \to 2.82$ from 2 to 40 steps). Accommodation biased $2.8\times$; shock $\mathrm{d}(\rho_2/\rho_1)/\mathrm{d}u_1$ **wrong sign**.
- VSS α: argon's $\alpha = 1.40$ sits at the stationary point $\sqrt{2}$ of $\langle \cos^2 \chi \rangle$; $\mathrm{d}T_y/\mathrm{d}\alpha \approx 0$ is the right answer (AD tracks FD, $|z| \leq 0.96$, across α).

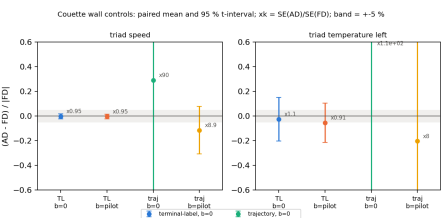


Score estimators against CRN FD (review §4.8, 384 seeds)

Couette ($N = 256$, 10 steps) and inlet shock ($N = 128$, 4 steps):

control	arm	AD − FD	z	SE(AD)/SE(FD)
Couette u_{wall}	terminal-label	-0.001 ± 0.011	-0.2	0.95
	trajectory, pilot baseline	-0.068 ± 0.11	-1.2	8.9
	trajectory, no baseline	$+0.17 \pm 1.2$	0.3	90
Couette T_{left}	terminal-label	-0.004 ± 0.03	-0.3	1.1
	trajectory, no baseline	$+1.19 \pm 2.4$	1.0	106
inlet T	terminal-label	$+0.8\,\%$	2.6	1.0
inlet n_ρ	every score arm	—	—	**185–1300**

- The trajectory score is unbiased where resolvable but its SE grows like $M^{2.8}$ in the horizon and $N^{0.56}$ in the particle count (it scores every event).
- A stochastically rounded count carries a score $\mp 1/f$: the reservoir density is out of reach of every score route. This decided the move to the ensemble adjoint (2026-09-15).



P3 replication and the two quadratures of the ensemble adjoint

P3 Test 4 (Couette walls), $N = 32\,000$, 32 seeds: $\mathrm{d}J/\mathrm{d}u_{\mathrm{wall}} = 3.032 \pm 0.010$ against the paper's 3.028 — a replication at the paper's precision.
P3 Test 3 (two-sided inflow): the transcription's own gradient is known to 1 % (1600-seed FD 0.1439 ± 0.0014); P3's retained-subset quadrature (a) depends on the retained count and on the reconstruction (affine $+18\,\%$, a Jensen bias of an affine fit to a convex field); the paper's 0.1517 is **3–5 %** above every estimate of the transcribed model: a forward-model difference at the inlet, not an adjoint one.
Route (b): values carried on the particles, realised pairs as the collision quadrature (one interpolated ghost per collision). On P3's model, two steps with strong collisions: interpolation-free (b) against CRN FD $+0.0008 \pm 0.0015$ (**0.5 %** resolution); route (a) with every particle retained $-0.8\,\%$ (3σ). $\mathrm{N}_0 = 16\,000$: (b) 0.1469 ± 0.0005 vs FD 0.1439 ± 0.0014 ($z = 2.0$) / 0.1466 vs 0.1453 ($z = 0.8$). Cost $1/40$ of (a).

On the production solver (VSS, NTC, move-then-collide, stochastic counts), inlet T, $N = 128$: the affine ghost basis read $-8.3\,\%$ ($z = -4.3$; collision term 27 % of the gradient); adding the $|v|^2$, x^2 columns (quadratic basis) gives $z = -0.95$. Reservoir density: **0.43× the FD's noise**, unbiased.

Walls as kernels, kernel parameters, and the exact sequential pull-back

Diffuse walls: the realised re-emission is a sample of the kernel, so wall T, u and accommodation enter only through the score of the re-emission density; no ε. **Kernel parameters** (`diam`, `omega`): $\partial_\theta \log q$ on the accepted pairs times their four-term.

The Δt-linearised pull-back is biased where particles collide more than once a step: relaxation `diameter` $-9.9\,\%$ ($z = -4.2$) at 0.16 collisions per particle-step. **The sequential pull-back** (accepted attempts in reverse slot order, ghosts from the field as it stood after each) is the exact discrete adjoint of NTC: -0.667 ± 0.039 vs FD -0.675 ± 0.017.

Every control, sequential pull-back (review §5.7), CRN FD $h = 0.1$:

control	ensemble	FD	z	SE ratio
Couette accommodation	0.2917 ± 0.009	0.2615 ± 0.017	1.7	0.52
Couette u_{wall} ($\varepsilon = 0$)	0.5865 ± 0.014	0.5933 ± 0.0093	-0.53	1.47
Couette T_{left}	0.1634 ± 0.0088	0.1804 ± 0.012	-1.2	0.73
inlet n_ρ, $N = 128$	$1.375\text{e}5 \pm 1.3\text{e}3$	$1.377\text{e}5 \pm 2.5\text{e}3$	-0.08	0.53
inlet T, $N = 256$	$1.364\text{e}5 \pm 3.6\text{e}3$	$1.374\text{e}5 \pm 1.1\text{e}3$	-0.32	3.3
relaxation `diameter`	-0.663 ± 0.026	-0.6723 ± 0.012	0.32	2.1
relaxation `omega`	0.2824 ± 0.010	0.2607 ± 0.014	1.2	0.69

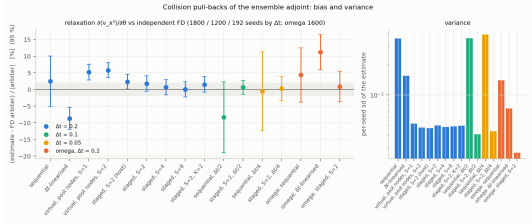
The accommodation FD is a step function of a decision parameter and cannot arbitrate below $\sim 3\,\%$.

Virtual-partner pull-back: bias and variance (review §5.12)

The sequential form's noise grows with the collisions per particle-step and blows up above ~ 0.5. P3's virtual-partner average, with NTC weights, all four values from one cell fit, S substeps. Relaxation `diameter` against a 1800-seed FD-only arbiter -0.6800 ± 0.0034:

pull-back	estimate	per-seed sd	bias
sequential	-0.663 ± 0.026	0.36	$+2.5\,\%$ (0.7σ)
Δt-linearised	-0.739 ± 0.011	0.15	$-8.6\,\%$ (5σ)
virtual, end-of-step nodes, $S = 2$	-0.641 ± 0.007	0.048	$+5.8\,\%$ (5σ)
virtual, staged nodes, $S = 2$	-0.668 ± 0.007	0.050	$+1.8\,\%$
virtual, staged, $S = 4$	-0.675 ± 0.007	0.048	$+0.7\,\%$
virtual, staged, $S = 8$	-0.680 ± 0.007	0.049	$+0.1\,\%$

- Residual $O(\Delta t/S)$; per-seed sd $7\times$ below sequential at 0.16 coll./step, 37–56$\times$ at 0.8.
- Staged nodes (the slot-order state at each substep) remove the frozen-field bias.
- Shipped: $(S, K) = (4, 8)$, staged.

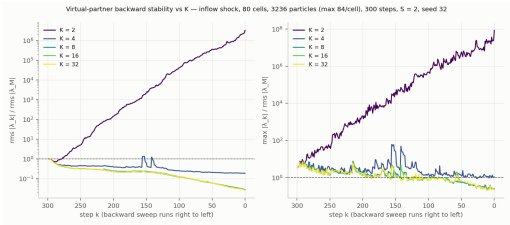


Partner count, stability, and the Rankine–Hugoniot acceptance case

Stability (300-step shock, ≤ 84 particles per cell): $K = 2$ grows geometrically (3×10^6 over the horizon); $K = 4$ marginal; $K \geq 8$ one decaying curve. The backward is refused when rms λ grows more than $100\times$.

Acceptance case $\mathrm{d}(\rho_2/\rho_1)/\mathrm{d}M_1$, 300 steps, where the pathwise derivative had the wrong sign: ensemble 2.74 ± 0.66, CRN FD 2.54 ± 0.17 / 2.17 ± 0.42, $z = 0.3$. Right sign; coarse (per-seed sd $4\times$ the FD's): long transients cost variance; prefer windowed, quasi-steady objectives.

Virtual pull-back, every control (staged, $S = 2$, $K = 8$): inlet T/n_ρ at 0.8 collisions per particle-step $z = -0.7/-0.4$, now below the CRN FD's own noise; Couette u_{wall} $z = +0.8$, T_{left} $+1.7$; `omega` $+0.9\,\%$ against a 1600-seed arbiter.



First shape oracles (2026-09-17/18)

Planar oracle (one step, collisionless, specular wall): analytic $\mathrm{d}J/\mathrm{dgap} = -0.5917$. The pathwise gradient at frozen hits reads **0** (the hit set is invisible to it); the value-weighted ε-hit score 0.588 ± 0.008.
G3a window score (1-D closed box, windowed impulse): the ε hit-timing score has no resolvable mean on a steady window and costs 4–8$\times$ the FD's sd $\Rightarrow$ the shape runs use $\varepsilon = 0$; the hit set enters through the replayed wall map and band terms instead.

Value-form rewards on Couette (48 seeds, CRN FD):

gate	objective, control	ensemble vs FD	z
G1	windowed v_y^2, wall speed	$+0.491 \pm 0.025$ vs $+0.4925 \pm 0.016$	-0.05
G2	windowed shear (per-hit), wall speed	-0.0177 ± 0.0014 vs -0.0165 ± 0.0008	-0.77
G2	same, T_{left}	-0.0025 ± 0.0006 vs -0.0032 ± 0.0015	$+0.46$

The first G2 read $40\times$ too small: the weight of a hit's kernel score must include the hit's own reward (reward-to-go from the draw onward).

G3: a translated wall in a 2-D Couette, and what it taught

1. **Hits change step.** Windowed v_y^2: pathwise alone $+0.82 \pm 0.23$ vs FD $+0.05 \pm 0.15$; the hit-timing band term (-0.43 ± 0.02) closes it.
2. **Two-sided fits are biased next to a wall.** Windowed shear came out with the wrong sign (-0.012 vs $+0.017$); nodes restricted to the leaving side ($v \cdot n > 0$) fix the sign.
3. **Bias or horizon?** With replica-exact value gradients the estimator equals the *truncated* FD on both objectives ($z = 0.5$); the gap to the full FD is the captured horizon (72 captured steps before the window close it).
4. **Estimators of $\nabla_x \lambda$** against replica-exact truth (windowed v_y^2, exact x/v part $+0.374 \pm 0.111$): one-sided -0.29 ± 0.17 bias, pooled -0.20 ± 0.11, one-sided + along-flight -0.08 ± 0.11, pooled + along-flight -0.07 ± 0.11. The along-flight identity holds per particle ($-1.3\text{e}{-5} \pm 2.8\text{e}{-5}$ on a scale $1.6\text{e}{-3}$).
5. **Unresolved**: the x/v part of a *translated flat wall* (a window-end delay sensitivity); every fitted or value-based form reads 0.

Decisions (2026-09-20): shape controls are normal displacements of curved surfaces; the body is a C^1 level set (a polygon's vertex term is a representation artefact).

Circle radius, level set and normal modes (2026-09-18 . . . 09-21)

Circle radius, analytic body, 48 seeds, collisionless:

term	10^{-20} N/m
tally	1.28 ± 0.01
$\nabla_v \lambda$ part	0.02 ± 0.02
$\nabla_x \lambda$ part	0.14 ± 0.09
ensemble	1.44 ± 0.09
CRN FD	1.395 ± 0.02
analytic F/r (no blockage)	1.23

$z = +0.5$. The tally reproduces F/r to 4 %; the rest is the blockage feedback.

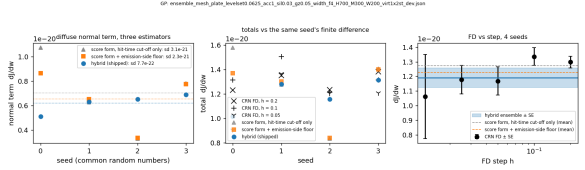
Level set: the same simulation as the analytic body to the surface error (objective to 2×10^{-6}); radius $z = -1.4$ (4 seeds).
$\cos 2\theta$ **mode** (coarse, 4 seeds): ensemble -1.587 ± 0.030 vs FD $-1.617/-1.571$ ($z = +0.7/-0.2$); per-seed sd 3.8 % (radius: 45 %).
Sphere radius (8 seeds, cluster): 3.389 ± 0.048 vs FD 3.487 ± 0.031 (10^{-20}), $z = -1.71$.
Lane vs stencil tracker: gradients agree to 2.3×10^{-15}.
First optimisation: 8 GD iterates, drag $-7.4\,\%$ at fixed area.

Grazing hits and the diffuse terms (2026-09-21/22)

Grazing regularisation: the per-hit derivative carries $1/(v \cdot n)$; one hit at $|v \cdot n|/|v| = 3 \times 10^{-3}$ was **15 %** of a window's tally. Floor in the derivative only:

ϵ	0	0.01	0.03	0.1	0.3
seed sd of the tally term	8.4 %	5.9 %	4.2 %	2.9 %	2.1 %
paired bias vs $\epsilon = 0$	—	$+1.0 \pm 1.1$	$+0.5 \pm 1.8$	-1.2 ± 2.3	-7.4 ± 2.4

Diffuse capsule plate (GP), control = half-width: first pass 80 % of FD (silhouette term needed; score resolution); then the **hybrid normal term** (pathwise reward through the reparameterised re-emission + score on the value after the hit) cut the normal term's seed sd $4\times$: 1.191 ± 0.068 **vs FD** 1.169 ± 0.096 ($z = +0.19$; $|z| \leq 1.6$ at five FD steps). Silhouette = **45 %** of the gradient.
Diffuse circle (GD): 1.348 ± 0.25 with timing vs FD 1.442 ± 0.03 / 1.128 ± 0.16; silhouette 61 %; 9 % low against the precise FD without the timing term (open then; see Part IV).



The specular capsule and the future-class fit (2026-09-22, superseded)

Specular capsule: mixed-pool fits read $-27\,\%$ ($z = -11$); the tally alone matched the FD seed by seed, and the deficit sat in the $\nabla\lambda$ parts. Cause: at the curved cap the one-sided rule does not separate leaving particles from those heading for the flat face.
Stratifying the fits by "a body hit at a later step" fixed it: $0.964 \pm 0.037 \to 1.356 \pm 0.011$ vs FD 1.35 ($z = +0.08$), and became the default.

It was biased by construction (found 2026-09-25, explained 2026-10-02): the class is a property of the particle's future, so the fit drops $\nabla P\,(\lambda_1 - \lambda_0)$ (Part IV, Finding 1). On the capsule $P \approx 0$ after a reflection, so it was harmless there; on the fine $\cos 2\theta$ mode (32 seeds, cluster) it read -2.386 ± 0.005 against FD -1.592 ($z \approx -120$), the tally alone matching the FD ($z = -0.33$).

Consequences: the `optc` optimisations ran with the class; the squash studies did not. The class is now a diagnostic option that warns.

Cluster studies: optc free-shape optimisation (2026-09-22 . . . 09-25)

Coarse circle, $f_{\mathrm{num}} \times 1$, specular, collisionless, fixed area; $S \in \{1, 2, 4, 8\}$ CRN seeds per iterate $\times$ {GD, Adam}; along-pool fits *with the future class*.

run	iterates	drag change (in-sample)	held-out (4 seeds)	end
S1 GD	131	$-14.9\,\%$	within 1 %	cancelled
S2 GD	216	$-19.2\,\%$	within 1 %	time limit
S4 GD	113	$-19.0\,\%$	within 1 %	time limit
S8 GD	61	$-13.9\,\%$	within 1 %	host OOM
S1 Adam	250	$-37.1\,\%$	within 1 %	complete
S2 / S4 / S8 Adam	96 / 102 / 37	$-38.8 / -35.0 / -41.8\,\%$	within 1 %	inflow face / limit

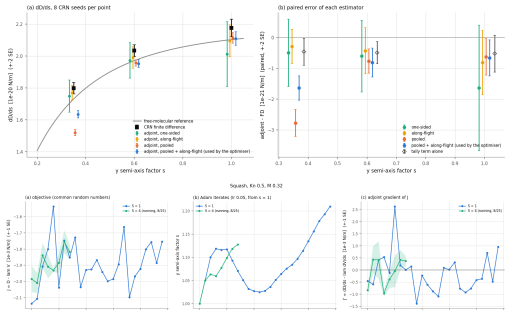
- Every failure on the way was a real tracking or meshing defect, fixed at its root (reinitialisation band, smoothed Adam step, stuck particles, concave re-entry, split cells, replay tolerance 10^{-9}), and led to the lane tracker ($2\times$ faster, $-45\,\%$ memory).
- The Adam bodies reach the outflow and inflow faces: their reductions are not those of an isolated body (a clearance constraint is needed). The gradient they followed was class-biased.

Cluster studies: the squash gate and optimisations (2026-10-01/02)

Free-molecular gradient gate ($f_{\mathrm{num}} \times 1$, 8 seeds), adjoint − CRN FD (% of FD):

variant	$s = 0.35$	0.6	1
tally alone	-2.6	-2.4	-2.4
one	-2.8	-3.0	-7.5
one_pool	$\mathbf{-15.5}$	-3.8	-2.9
along	-1.6	-2.1	-3.8
along_pool (used)	$\mathbf{-9.1}$	-4.0	-3.0

($\pm$1–5 %; along_pool $z = -8.3$ at $s = 0.35$.)
Optimisations (Adam, 25 iterates, along_pool, 0.2-step timing band, no cut-volume or order terms, identity without $\mathcal{L}^*\lambda$): free-molecular S1/S4: $s = 0.355/0.339$ (reference optimum 0.35); Kn 0.5: S1 drifts to $s = 1.21$, S4 to 1.13 (8 iterates); Kn 0.5, M 10: S1 to 0.48 (12 iterates).



Bias study (2026-10-02 . . . 10-04): method

Small runs on the 4 GB workstation GPU; every estimator paired with the CRN FD of the same seed (`scratch/bias/`: `kn_gate.py`, `fm_exact.py`, `horizon.py`, `fd_only.py`, `summarise.py`).

- Squash ellipse ($R = 1.5$, $h_\phi = 0.0625$, grazing 0.05), 20×20 cells, warm-up 700, 300 captured steps, window 200, CRN FD $h = 0.05$.
- Free-molecular and Kn $= 0.5$ ($\nu \Delta t \approx 2.5\,\%$) at $f_{\mathrm{num}} \times 2$ (75 particles per cell); virtual pull-back 2×4 locally.
- All $\nabla_x \lambda$ variants from one replay, so their differences are paired too.

An exact free-molecular reference that failed: without collisions a path is a deterministic function of θ, so the forward-mode JVP chained step to step is the exact pathwise derivative (reproduces the drag to 2×10^{-16}). The box is a dispersing billiard (convex body, specular walls): tangents grow at every bounce, and the chained JVP ranged from $-5.5\times$ to $+8.8\times$ the tally term over 8 seeds. Useless per seed; the fits' smoothing is what makes the ensemble adjoint usable.

Audit: every θ-dependence of the forward

dependence	term	before the study	now
hit times, points, specular v^{out}	pathwise x, v	in	in
reward through v^{out}	tally	in	in
diffuse kernel normal	score	in	in (0 specular)
hit $\leftrightarrow$ miss	silhouette	in	in (off for specular)
hit crosses a window edge	timing	**reported, not added**	in, 10-step band (F6)
hit/collision order at every step	order	**missing**	in (F7)
cut-cell volume in NTC	operator	**missing**	in (F2)
$\nabla_x \lambda$ along v with collisions	along-flight	$\mathcal{L}^* \lambda$ **dropped**	in (F3)
node sets of the fits	—	**future class (optc); pool across body**	state-based (F1, F5)
newborn / leaving particles' hits	—	unmonitored	warns above 10^{-3} (F4)

The VJP of the move is complete: every floating output of the wall map gets a cotangent, and the replay is checked against the record at every step. F1–F7: the findings that follow.

Finding 1: the future-class fit is biased, and cannot be repaired

Coarse circle, level-set body, `mode2` ($\cos 2\theta$), 4 CRN seeds (32–35), the same records for every row; FD $= -1.594 \pm 0.043$, tally term -1.596 ($\times 10^{-20}$), so the $\nabla\lambda$ parts should sum to FD−tally $= +0.01 \pm 0.68 \times 10^{-21}$.

fit class	x part 10^{-21}	v part 10^{-21}	gradient 10^{-20}	gradient − FD (paired) 10^{-20}
no class (`none`)	-0.28	$+0.36 \pm 0.15$	-1.587 ± 0.030	$+0.007 \pm 0.071$
hit-ahead, current	-0.72	-6.02 ± 0.34	-2.269 ± 0.036	-0.675 ± 0.057
hit-ahead, stopped at the next birth	-0.80	-10.82 ± 0.62	-2.758 ± 0.069	-1.164 ± 0.091

Refilling is not the cause. A purer class makes the bias *larger*. The stratified fit estimates $\nabla\lambda_\kappa$ within each class, while

$$\lambda = P\lambda_1 + (1-P)\lambda_0 \;\Rightarrow\; \nabla\lambda = P\nabla\lambda_1 + (1-P)\nabla\lambda_0 + \underbrace{\nabla P\,(\lambda_1 - \lambda_0)}_{\text{dropped}},$$

with $P(z)$ = probability of a later body hit. A reflected particle can come back, most likely after a specular y-wall bounce (not checked), so $\nabla_v P \neq 0$, and $\lambda_1 - \lambda_0 \approx$ the next hit's drag. Neither the silhouette nor the timing term carries this. On the capsule, $P \approx 0$ after a reflection, so the class was harmless there.

Not repairable: restoring $\nabla P(\lambda_1 - \lambda_0)$ needs ∇P from the same local regression, and that sum is the mixed-pool fit again, with more variance. Free-molecular squash: $+4$ to $+20\,\%$. Valid stratification uses the present state ($v \cdot n$, the tangent half-space).

Free-molecular specular squash: adjoint − FD, % of FD

variant	$s = 0.35$		$s = 0.6$		$s = 1$	
	no timing	+ timing	no timing	+ timing	no timing	+ timing
`one`	-3.0 ± 0.8	-2.1 ± 1.2	$+1.3 \pm 4.1$	$+1.3 \pm 3.9$	-1.8 ± 2.3	-1.7 ± 2.3
`one_pool`	-19.4 ± 0.7	-18.4 ± 1.1	-5.1 ± 1.3	-5.1 ± 1.8	-2.3 ± 1.5	-2.3 ± 1.6
`along`	-1.7 ± 1.2	-0.7 ± 1.7	-0.7 ± 2.4	-0.7 ± 2.4	-0.9 ± 1.5	-0.9 ± 1.5
`along_pool`	-9.9 ± 1.3	-9.0 ± 1.7	-4.5 ± 1.4	-4.5 ± 1.9	-2.1 ± 1.4	-2.1 ± 1.6
`one_pool_plane`	-2.7 ± 0.6	-1.7 ± 0.9	-2.6 ± 1.2	-2.6 ± 1.9	-1.5 ± 1.5	-1.5 ± 1.6
`along_pool_plane`	-1.7 ± 1.1	-0.7 ± 1.5	-2.5 ± 1.4	-2.5 ± 1.9	-1.5 ± 1.4	-1.4 ± 1.5

8 CRN seeds per s ($f_{\mathrm{num}} \times 2$), ± 1 SE of the paired difference. "+ timing": window-edge term (0.2-step band at $s = 0.35, 0.6$; 10-step band at $s = 1$).

- `one_pool`, `along_pool` (the cluster runs' variant): $-19\,\%$ / $-9\,\%$ at $s = 0.35$, $-5\,\%$ at $s = 0.6$ — Finding 5.
- Restricted to the hit's half-space (`*_plane`): every entry within 2σ; pooled over s a residual of about $-2\,\%$ remains (located below).

Finding 5: the pooled fit reaches across a thin body

The pooled fit draws its $k = 96$ nodes from the 3×3 block around the hit's cell (1.5×1.5 m). At $s = 0.35$ the body is $2sR = 1.05$ m thick, two cells: the block around a hit on the upper face holds the particles *below* the body. Their values have nothing to do with the query's, and the one-sided rule does not separate them (below the body, $v \cdot n_{\text{top}} > 0$ means moving towards the lower face).

Fix: drop the nodes more than a quarter cell behind the plane through the query, normal to its hit normal, $(x_{\text{node}} - x_q) \cdot n \geq -\frac{1}{4}\Delta x$. Variants `one_pool_plane`, `along_pool_plane`; the latter is the default everywhere. It also removes most of the v part's spread ($-1.1 \to -0.12 \times 10^{-21}$ at $s = 0.35$).

Finding 6: the window-edge timing term

- It was left out as "zero mean on a steady window". The window is steady (drag change over the 200 steps $-0.1\,\%$ specular, $+0.8\,\%$ diffuse).
- The 0.2-step band (one step per edge, ∼10 hits per seed) read $+2.8 \pm 1.2\,\%$ at $s = 1$ and $-35\,\%$ on one diffuse seed: first mistaken for the missing $-2.4\,\%$; it is noise.
- 10-step band (whole steps per edge), same 8 seeds: $+0.03 \pm 0.25\,\%$.

Decision: the term belongs to the windowed objective's derivative (nonzero whenever the window is not steady), so it is in the reported gradient, with the 10-step band as default.

Diffuse squash ($a = 1$, free-molecular, $s = 1$, 8 seeds, silhouette band 0.03):

variant	$s = 1$	
	no timing	+ timing
`one`	$+1.0 \pm 3.1$	$+1.2 \pm 3.0$
`one_pool`	-6.0 ± 3.1	-5.8 ± 3.0
`along`	-0.6 ± 3.4	-0.4 ± 3.3
`along_pool`	-3.2 ± 3.4	-3.1 ± 3.3
`one_pool_plane`	-3.1 ± 3.1	-2.9 ± 3.0
`along_pool_plane`	-1.6 ± 3.4	-1.4 ± 3.3

9.12 (tally, hybrid) $+ 9.84$ (silhouette) $- 0.13$ (normal score) $- 0.15$ ($\nabla\lambda$) $[10^{-21}]$ against FD 19.0: the silhouette term is half the gradient; every variant but the cross-body `one_pool` within its $\pm 3\,\%$ noise. The first diffuse squash gate.

Finding 2: the cut-cell volume term, built and gated

NTC's $1/V_c$ was seen by neither the replay nor the VJP. Built 2026-10-02 (derivation: Term 7).
Machinery gate (Kn $= 0.5$, $f_{\mathrm{num}} \times 4$, 8 seeds): scale the NTC volume of a set of cells by $1+\epsilon$ — the only change — and compare the FD in ϵ with the adjoint's $\sum_c g_c V_c$:

cells scaled	FD	adjoint	z
all	2.11 ± 0.26	2.45 ± 0.08	-1.2
cut cells	0.69 ± 0.45	0.34 ± 0.01	$+0.8$

(10^{-3} N/m)

Geometry factor: $\sum_c \partial V_c/\partial s = -7.068$ against $-\pi R^2 = -7.069$.
Size at Kn 0.5, $s = 1$: $-4.4 \times 10^{-4} = -2.9\,\%$ of $\mathrm{d}D/\mathrm{d}s$ (seed SE 0.1 %).

Finding 3: the along-flight identity with collisions

The identity used was $v \cdot \nabla_x \lambda = -T$; with collisions it is $-T - \mathcal{L}^* \lambda$ (the omission was justified at $\leq 0.3\,\%$ collisions per step; Kn 0.5 has 2.5 %). Fitted slope against the identity's value of its component along v, paired (x part, 10^{-4} N/m, 8 seeds):

$$\text{fit} - \text{identity}(T + \mathcal{L}^* \lambda) = -1.3 \pm 0.5, \qquad \text{fit} - \text{identity}(T) = +4.9 \pm 0.5.$$

- Without $\mathcal{L}^* \lambda$ (the cluster runs) the identity is off by 3.2 % of $\mathrm{d}D/\mathrm{d}s$ ($z \approx 10$); with it fit and identity nearly agree; the residual $-0.8\,\%$ is the $O(\nu \Delta t)$ of the discrete scheme.

Kn 0.5, adjoint − paired FD (% of FD; the $h = 0.05$ FD has 18 % per-seed noise):

variant	$s = 1$	
	no timing	+ timing
`one`	-7.3 ± 6.1	-7.2 ± 6.2
`one_pool_plane`	-8.0 ± 6.0	-7.8 ± 6.1
`along`	-6.2 ± 5.9	-6.0 ± 6.0
`along_free`	-9.9 ± 5.9	-9.8 ± 6.0
`along_pool_plane`	-7.2 ± 6.1	-7.1 ± 6.2
`along_pool_plane_free`	-11.0 ± 6.1	-10.8 ± 6.1

(`_free`: identity without $\mathcal{L}^* \lambda$.)

Finding 7: the order of a hit and the step's collisions

NTC collides once per step, after the move. A hit that slips from step k into $k+1$ collides in step k as the incoming particle instead of the reflected one: $J_c = c(x_w, v^{\mathrm{in}}) - c(x_w, v^{\mathrm{out}})$, $c = \Delta t\, \mathcal{L}^* \lambda$, at every step boundary. The window-edge reward jump is the collisionless remainder of the same event.

Kn 0.5, $s = 1$, 8 seeds: $+6.3 \pm 4.4 \times 10^{-5} = +0.4 \pm 0.3\,\%$ of $\mathrm{d}D/\mathrm{d}s$. Small, because the incoming and reflected collision effects differ by a small fraction of the reward; in the gradient by default.

Finding 4 (unscored hits): hits of particles born in the step or leaving the box in it get no reward in λ and no VJP. Their share is recorded; zero while the body is more than $|v|\Delta t$ from the inflow and outflow faces; nonzero only for the `optc` Adam bodies that reached those faces. Now warns above 10^{-3}.

Audit: where could a no-collision assumption hide?

The step is move (hits) → NTC collisions. Each term's justification, re-checked with collisions:

term	assumption	with collisions
pathwise x, v	$\nabla\lambda_k$ at the post-move state; λ_k includes step k's collisions	holds
tally, score, hybrid normal	exact reward; weights λ + later rewards	holds
silhouette	straight flight to the step end	holds (collisions follow the move)
timing band	a hit changing step matters only at the window edges	**wrong**: the hit/collision order swaps — Finding 7
along-flight	$v \cdot \nabla_x \lambda = -T - \mathcal{L}^* \lambda$	$O(\nu\Delta t)$ in the discrete scheme (F3)
cut-cell volume	—	built (F2)
cell membership of a hit particle	omitted (integer)	$O(\nu\Delta t)$ below the order term
fits at hit slots only	only hit slots' post-move states depend on θ	holds

The free-molecular $-2\,\%$: noise or bias?

Adjoint $-$ FD (%), 8 seeds per s, $f_{\mathrm{num}} \times 2$; pooled with inverse-variance weights:

	$s = 0.35$	0.6	1	pooled
`along_pool_plane`	-1.7	-2.5	-1.5	-1.8 ± 0.7 ($z = -2.5$)
`one_pool_plane`	-2.7	-2.6	-1.5	-2.6 ± 0.5 ($z = -5.3$)
tally term alone	-1.2	-2.6	-1.2	
tally alone, $f_{\mathrm{num}} \times 1$ (cluster gate)	-2.6	-2.4	-2.4	(± 0.9–1.4 each)

Systematic, and it sits in the *tally* term, which is exact given the hits.

- FD curvature ($h = 0.05$, on the analytic reference): $\leq 0.15\,\%$, wrong sign.
- Grazing cut-off removed (same forward): $+0.15 \pm 0.25\,\%$.
- **The captured horizon**: particles reflected *before* the captured steps bounce off the specular y walls ($\sim$230-step round trip) and re-hit inside the window; the adjoint cannot see their first hit's dependence on the shape. Capturing 600 instead of 300 steps (same forward, $f_{\mathrm{num}} \times 4$, 8 seeds): $+0.8 \pm 0.25\,\%$ at $s = 1$, $+1.1 \pm 0.3\,\%$ at $s = 0.35$.

Most of the $-2\,\%$ is truncation, not a code error; it grows with the box's memory (reflective walls, trapping bodies). Gates at the 1 % level: ≥ 600 captured steps or open y faces. **Cluster** (`fm_hor_s*`, $f_{\mathrm{num}} \times 1$, 8 seeds, the whole 1000-step run captured): $+0.3 \pm 0.2\,\%$ ($s = 1$), $+0.6 \pm 0.1\,\%$ ($s = 0.35$); the whole run still reads -1.0 ± 1.2 / $-1.8 \pm 0.6\,\%$: at this density the horizon is a third of the remainder.

Kn $= 0.5$: the remaining gap, and what it is not

Precise FD (FD-only, 32 seeds, $f_{\mathrm{num}} \times 2$): $h = 0.1$: 0.01650 ± 0.00019; $h = 0.2$: 0.01605 ± 0.00010; Richardson $\mathbf{0.01665 \pm 0.00026}$.
Adjoint (8 seeds, every term): `along` $-6.5\,\%$, `along_pool_plane` $-7.5\,\%$, `one_pool_plane` $-8.3\,\%$ (each ± 1.8); the old configuration $-9.7\,\%$.

Ruled out (paired, same forward):

- horizon 300 $\rightarrow$ 450 captured steps: $+0.1 \pm 0.4\,\%$ (fit variants);
- virtual quadrature $S \times K = 2 \times 4 \rightarrow 4 \times 8$: $< 1\,\%$;
- fit width $k = 48, 96, 192$ pooled: within 1 % (unpooled $k = 16$: $+4$–$10\,\%$, noisy);
- one-sided fits in the backward next to the body: $< 1\,\%$; timing band: $+0.15\,\%$;
- the sequential (exact) pull-back as referee: $50\times$ noisier at this collision rate — useless.

Update 2026-10-12 (cluster, $f_{\mathrm{num}} \times 1$, section V): the 64-seed FD has no curvature ($h = 0.1$ vs 0.2: $+0.14 \pm 0.67\,\%$), so the Richardson value above was noise; against it the gap is $-3.7 \pm 0.8\,\%$, the same for the per-query and per-cell fits. The whole-run horizon ($+2.8 \pm 0.8\,\%$, 4 seeds) and a cubic-in-v fit ($+3.1 \pm 0.2\,\%$) are each of its size.

What the week's changes do to the squashkn result

The squashkn cluster runs (2026-10-01/02) used the then-default estimator. Same 8 Kn 0.5 seeds, $s = 1$, paired, % of the old gradient:

change	effect on $\mathrm{d}D/\mathrm{d}s$	sign of the old error
$\mathcal{L}^*\lambda$ in the along-flight identity (F3)	$+4.1 \pm 0.2\,\%$	too low
half-space restriction of the pooled fit (F5)	$+0.8 \pm 0.05\,\%$	too low
cut-cell volume term (F2)	$-2.9 \pm 0.1\,\%$	too high
hit/collision order term (F7)	$+0.4 \pm 0.3\,\%$	too low
10-step timing band (F6)	≈ 0 in the mean; far less noise	—
net	$+\mathbf{2.4}\,\%$ ($+2.0 \pm 0.16$ without F7)	

Against the precise FD: old $-9.7\,\%$ $\rightarrow$ now $-7.5\,\%$ (the open gap).

On the optimum: $J'(s) = \mathrm{d}D/\mathrm{d}s - \lambda_A\,\mathrm{d}A/\mathrm{d}s$ is the difference of two nearly equal numbers ($\lambda_A\,\mathrm{d}A/\mathrm{d}s = 1.537 \times 10^{-2}$, $\mathrm{d}D/\mathrm{d}s \approx 1.47 \times 10^{-2}$ at $s = 1$), so a few per cent of $\mathrm{d}D/\mathrm{d}s$ moves the zero of J' a long way. A linear fit of the old runs' $\mathrm{d}D/\mathrm{d}s$ in s ($\pm 50\,\%$ on the slope) puts the old zero at $s^* \approx 1.08$–1.11, the corrected one at ≈ 1.04–1.05, and one matching the FD ($+8\,\%$) at ≈ 0.92–0.94. The Kn 0.5 optimum is ill-conditioned; re-run squashkn only after the gap is closed. M 10: the corrections were not measured there.

Why the interpolant was reopened (2026-10-09/10)

The virtual pull-back asks $(3+2K)N$ field values per substep ($19N$ at 4×8): the node's own value, the two post-collision ghosts of each virtual partner, the boundary move. Until 2026-10-11 each was a **per-query** weighted quadratic fit over the k nearest nodes of the query's cell.

Its answer depended on k (Kn 0.5 squash, $s=1$, $f_{\mathrm{num}}\times 4$, paired, % of $\mathrm{d}D/\mathrm{d}s$):

k	16	32	48	64	96, 128
vs $k=128$ (3 seeds)	diverges	-3.2 ± 0.3	0.0 ± 0.7	$+0.1\pm 0.1$	0 (whole cell)
seconds per seed	465	670	790	985	1295, 1660

Over 12 seeds: $k=32$ $-3.5\pm 1.8\,\%$ vs FD, $k=64$ $-2.0\pm 2.0\,\%$ (k 32→64: $+1.6\pm 0.7\,\%$, half the 3-seed estimate). Cost: ≈ 300 s fixed $+\ 10.7$ s per unit of k; at $k\geq$ occupancy the fit is the whole cell and its weights $1/(1+(d/d_k)^2)$ only span $[\frac{1}{2},1]$.

Idea: if the fit spans the cell anyway, fit *once per cell* (C fits, not $19N$) and evaluate the polynomial at every query. Settled first: S, K (S1/S2/S8, K4/K16 within $\pm 3\,\%$, flat at 4×8) and the simpler collision adjoints (23 stored gates reproduce to 7×10^{-10}).

The per-cell interpolants tried

One least-squares polynomial per collision cell in scaled offsets $o = (\tilde{p} - \bar{p}_c)$ from the cell's node mean ($\tilde{p} = (x/\Delta x, v/v_s)$), relative ridge 10^{-8}; `cell_fit.py`, `basis="cell:<poly>[:opt]"`.

basis	what it is
`mean`, `lin`	constant (1 column); $1, o$ (6 in 2-D)
`quad`	the per-query fit's design: $1, o, v \otimes v, x_d^2$ (14)
`quadfull` / `cubicv` / `cubic`	every product (21) / + cubic in v (24) / every monomial to order 3 (56)
`adapt`a	**order guard**: highest order whose cumulative columns $\leq$ count$/a$; else a plane, else the mean
`loo`	per cell the order (0..max) of least leave-one-out residual $\sum w\,(r_i/(1-h_{ii}))^2$
`pool`r	fit the $(2r+1)^D$ block, tricube weight in x (support $r+1$ cells), evaluate in the cell
`rb`	partner ghosts averaged over the VSS angle: 6 sigma points, exact for quadratics (any α)
split (normals)	three fits per cell: leaving / approaching the surface ($v \cdot n_c \gtrless 0$) / all
`align`r, `both`	cells with $\lvert\sum n_i\rvert/\#n_i < r$ (complex): per-query fallback, or two-sided (`both`)

The per-query fit's weights cannot be reproduced per cell (they are centred on the query); the per-cell fit is unweighted (or x-weighted when pooled).

Correctness of the machinery

- every basis reproduces its own polynomials to 10^{-15}; the fit is the numpy least-squares solution to 10^{-13}; the slopes are the exact polynomial derivative (4×10^{-11});
- order guard: 60, 40, 25 nodes $\rightarrow$ quadratic, 14 $\rightarrow$ plane, 3 $\rightarrow$ mean; leave-one-out picks the mean on pure noise and the quadratic on a quadratic field;
- sigma points against 4×10^5 VSS scatterings: quadratics within 1.3 se at $\alpha = 1, 1.4$; cubics only at $\alpha = 1$ (67 se off at 1.4, as derived);
- split fits in **1, 2 and 3-D** (collide-only box, 3^D cells): with every cell complex the hybrid backward *is* the per-query one-sided backward (relative difference 0); fallback capacity = the measured load;
- an empty side (no node leaving the surface) gave NaN on a real geometry: now the identity, deferring to the unsplit fit (`test_cell_fit.py`, 12 tests).

0-D relaxation (t_x, 24 seeds, CRN FD): `cell:quad` $+0.0 \pm 3.0\,\%$ vs FD, $+0.3 \pm 0.7\,\%$ paired vs $k = 32$; backward 1.2 s vs ≈ 1100 s (CPU). A `lin` fit gives ≈ 0: $1, v$ are collision invariants, so a linear field has no collision term (the machinery's own check).

Kn 0.5 squash, 12 seeds: the decisive comparison

$s = 1$, $f_{\mathrm{num}} \times 4$, 4×8, CRN FD $h = 0.2$ per seed: 0.01597 ± 0.00026.

backward field	$\mathrm{d}D/\mathrm{d}s$	vs FD	$-$ (k=64): sd (max)	s / seed, bwd
per-query $k = 32$	0.01541 ± 0.00025	$-3.5 \pm 1.8\,\%$	2.5 (6.0) %	670 / ≈550 s
per-query $k = 64$	0.01565 ± 0.00030	$-2.0 \pm 2.0\,\%$	—	985 / 897 s
`cell:quad`	0.01665 ± 0.00107	$+4.3 \pm 7.9\,\%$	**28 (95) %**	98 / 13 s
`cell:quad:rb`	0.01544 ± 0.00057	$-3.3 \pm 4.0\,\%$	11 (34) %	112 / 27 s
`cell:quad:adapt1.5`	$\mathbf{0.01566 \pm 0.00024}$	$\mathbf{-1.9 \pm 2.3}\,\%$	5.3 (13) %	**95 / 13 s**
`cell:cubicv:adapt1.5`	0.01611 ± 0.00019	$+0.9 \pm 1.8\,\%$	5.5 (15) %	103 / 20 s
`cell:quad:pool1` (3 seeds)	0.01452 ± 0.00030	$(-10.9 \pm 4.4\,\%)$	3.8 (8) %	131 / 53 s

- **Unguarded** `quad` **is unstable**: seed 11 $+97\,\%$, seed 10 $-26\,\%$ (the per-query fits: $+2$, $-8\,\%$). Sparse cells' underdetermined quadratics answer the partner ghosts with extrapolated noise and the recursion compounds it. `rb` (seed 10 $-40\,\%$) does not cure it.
- **The order guard does**: `adapt1.5` matches $k = 64$ ($+0.05 \pm 1.5\,\%$) and the FD at $\frac{1}{10}$ of the seed's time; its seed-to-seed SE equals the per-query fit's.

Per seed: adjoint / FD -1, %

seed	FD	$k = 32$	$k = 64$	quad	quad:rb	quad:adapt1.5	cubicv:adapt1.5
0	0.01532	-9.5	-5.8	-2.8	-0.7	-5.0	-3.2
1	0.01744	-13.4	-11.0	-10.6	-10.0	-17.7	-7.9
2	0.01610	-7.7	-4.4	-3.5	-3.3	-5.4	-1.8
3	0.01629	$+1.8$	$+3.2$	$+2.7$	$+0.9$	$+3.6$	$+5.4$
4	0.01636	$+0.7$	$+3.2$	$+3.7$	$+3.1$	$+1.5$	$+4.9$
5	0.01547	-0.3	$+3.0$	$+3.4$	$+5.2$	$+1.2$	$+4.8$
6	0.01596	-5.1	-7.1	-7.0	-6.3	-7.8	-1.7
7	0.01598	$+3.0$	$+9.5$	$+13.8$	$+14.3$	$+2.9$	-0.3
8	0.01536	$+5.2$	$+4.7$	$+4.6$	$+5.0$	$+3.9$	$+6.2$
9	0.01686	-9.2	-9.9	-10.1	-10.5	-5.1	-3.0
10	0.01655	-8.0	-8.3	$\mathbf{-25.5}$	$\mathbf{-39.8}$	-5.5	-5.7
11	0.01393	$+2.3$	$+1.2$	$\mathbf{+97.2}$	$+6.2$	$+14.2$	$+16.5$

The per-seed spread is mostly the FD's own (CRN FD at $h = 0.2$, $\approx \pm 6\,\%$ per seed); the paired columns move together except where an unguarded fit breaks.

Variance: what `rb` buys, and why value cross-validation misleads

The backward's own noise (0-D relaxation, 8 forwards × 8 partner keys, sd of the gradient):

	sampled angle	`rb` (exact angle average)
4×8	1.37 %	0.91 %
4×2	2.95 %	1.85 %

Variance $/2.2$ — real, but small against a forward's seed-to-seed noise ($\approx 3.5\,\%$), and no help against the sparse-cell instability.

5-fold cross-validation of node values (squash snapshot, $k = 64$ backward, 3 steps): the values are nearly pure noise — the cell mean predicts held-out nodes at the noise floor (mse 3.0×10^{-12}, variance 3.1×10^{-12}); every unguarded quadratic overfits to $3\times$ ($k = 32$: 9.7, $k = 64$: 8.9, per-cell `quad`: 9.3, all $\times 10^{-12}$), mostly in cells of < 20 nodes; guarded / pooled / LOO fits reach 2.2–3.5. **But this cannot rank fits for the backward**: the cell mean wins it and has *no collision term* ($1, v, |v|^2$ are invariants; only the rest of λ enters $\mathcal{L}^*\lambda$). Only gradients against FD rank them.

The stored collisional gates, re-run with the per-cell fits

Paired with the gates' own per-query $k = 32$ runs (same forwards; `scratch/cellfit/gates*.jsonl`).

gate (seeds)	`cell:quad` vs $k = 32$	`adapt1.5`	`rb`
0-D relaxation t_x (24)	$+0.3 \pm 0.7\,\%$	—	$+0.3\,\%$ vs FD
Couette speed: energy, shear windows (4; 32)	$\leq 0.2 \pm 0.6\,\%$	same	same
Couette speed, one-sided split (2; 32)	$+0.2 \pm 0.6$, $+0.2 \pm 0.2\,\%$	same	same
Couette wall temperature (32)	$+0.3 \pm 1.3\,\%$	same	$+0.5 \pm 1.3\,\%$
Couette temperature, one-sided (2; 32)	$+4.5 \pm 7.0$, $+5.0 \pm 5.4\,\%$	same	same
inlet n / T, $N = 128$ (32)	$+0.7 \pm 0.3$ / $+2.0 \pm 1.1\,\%$	$+0.8 \pm 0.6$ / —	$+0.3$ / $+1.5\,\%$
relaxation `diameter` / `omega` (8)	$+1.2 \pm 0.3$ / $-0.3 \pm 0.6\,\%$	same	—

These cells hold $\leq\approx 40$ nodes (the order guard rarely engages); the FD noise of these gates (± 6 to $\pm 280\,\%$) is far larger than every difference. Each per-cell run took seconds where the per-query fit took up to 40 minutes on the CPU.

Cut cells on real geometry: survey and the one-sided failure

Geometry survey (optc iterates, normal resultant R per cell over its flow region): cells with $R < 0.9$ hold $\leq 6\,\%$ of the flow (circle: 0), $R < 0.8 \leq 5\,\%$; up to 24 cells contain opposed normals.

optc S1-adam iterate 100, Kn 0.5 ($f_{\mathrm{num}} \times 4$; $g \cdot \mathbf{1}$ = derivative along the uniform level-set shift; FD $h = 0.05$):

backward field	seed 1	seed 2
FD	-0.0046	-0.0055
per-query, one-sided $k = 32$ (the old option)	$+0.113$ (λ rms $\times 150$)	—
per-query, two-sided $k = 32$	-0.0024	$+0.018$
`cell:quad`, two-sided (unguarded)	-0.0028	-0.027
`cell:quad:adapt1.5`, two-sided	$\mathbf{-0.0049}$	$\mathbf{-0.0044}$
`cell:quad:adapt1.5`, split everywhere	-0.0040	$+0.0033$
`cell:quad:adapt1.5:align0.9:both` (shipped)	$\mathbf{-0.0035}$	$\mathbf{-0.0038}$

The circle at the same settings: every fit sane. Falling back to the per-query one-sided fit in complex cells inherits its divergence; the shipped choice leaves those cells two-sided. Two seeds of one shape: evidence, not a gate — the cluster `cellgate` runs six shapes × 16 seeds.

Selection

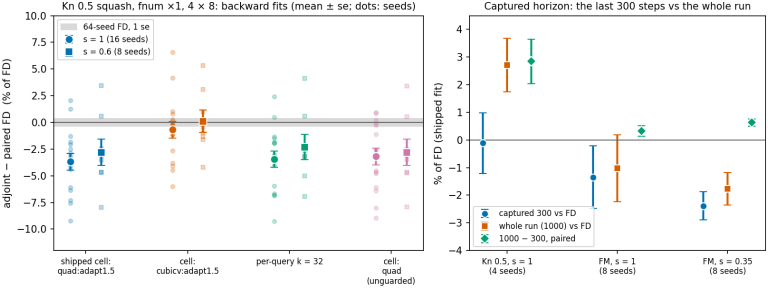
Shipped (`ensemble_backward.CELL_BASIS`, 2026-10-12): `cell:cubicv:adapt1.5:align0.9:both` (2026-10-11: `cell:quad:...`; the cubic-in-v columns are the only change, next frames)

- one quadratic (cubic in v) per cell, its columns capped at count/1.5 (else plane, else mean);
- with surface normals: split at $v \cdot n_c = 0$ (each side guarded), empty sides and cells with $R < 0.9$ two-sided;
- matches the 64-nearest fit ($+0.05 \pm 1.5\,\%$) and, at $f_{\mathrm{num}} \times 1$, the per-query $k = 32$ (-3.7 vs $-3.4 \pm 0.8\,\%$ against FD); seed-to-seed SE unchanged; every stored gate; stable on all six optc shapes $\times$ 16 seeds, where the per-query one-sided fit failed;
- **a squash seed** $985 \to 95$ **s** (backward $897 \to 13$ s): $\sim 10\times$ more optimisation iterates per GPU-hour.

Rejected: unguarded `quad`, `rb` (heavy tails); `pool1` (reads low, 3 seeds); the per-query fallback (unstable one-sided fit). **Kept as options**: `cell:quad:adapt1.5:align0.9:both` (the 10-11 default), `rb`, `loo`, `pool`, `own`, `quadratic_full`. The shape gradient's *slope* fits (pooled, one-sided, $k = 96$) are unchanged.

Cluster gates at production density (2026-10-12)

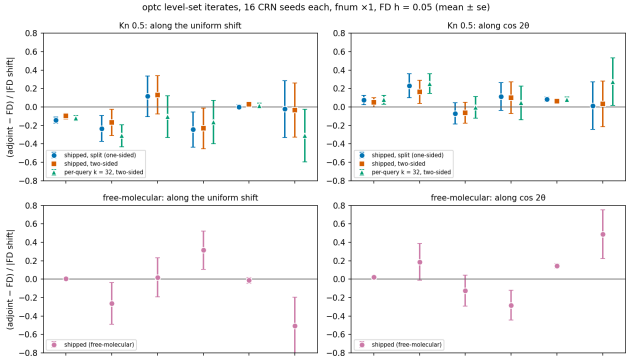
`cellgate` / `biasgate`, $f_{\mathrm{num}} \times 1$, 4×8, Kn 0.5 squash; every fit paired on the same forward and against the seed's CRN FD ($h = 0.2$); `scratch/cellfit/cluster_analysis.py`.



- **FD** (64 seeds): $h = 0.1$ / 0.2: 0.01633 ± 0.00010 / 0.01631 ± 0.00006, paired difference $+0.14 \pm 0.67\,\%$.
- $s = 1$ / 0.6: shipped -3.7 ± 0.8 / $-2.8 \pm 1.2\,\%$; per-query $k = 32$ -3.4 / -2.3; unguarded `quad` -3.2 / -2.8 (stable at this density); `cubicv:adapt1.5` -0.7 ± 0.8 / $+0.1 \pm 1.1\,\%$ (paired $+3.1 \pm 0.2$ / $+3.0 \pm 0.3\,\%$ over the shipped fit; 50 vs 46 s a seed; the per-query fit 1100 s).
- **Horizon**: the whole run captured adds $+2.8 \pm 0.8\,\%$ at Kn 0.5 (4 seeds), $+0.3$ / $+0.6\,\%$ free-molecular. Basis order and horizon are each the gap's size; additivity: `gapgate`, below.

Cluster gates: the optc shapes (16 seeds each)

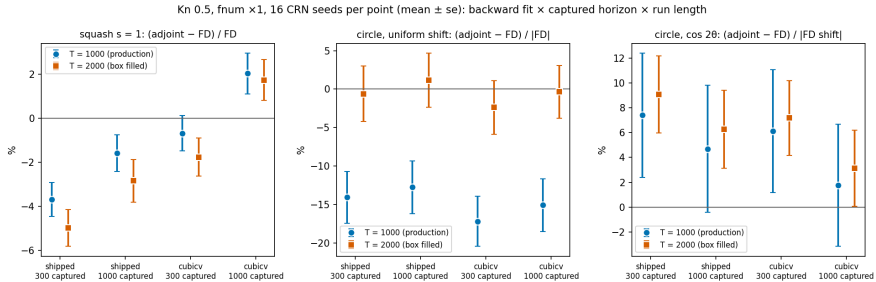
$g \cdot d$ vs CRN FD ($h = 0.05$) along the uniform shift and $\cos 2\theta$, in units of |FD along the shift|:



- **Stable**: every per-cell setting on every shape (λ rms $\leq 3.2 \times 10^{-7}$); the per-query one-sided fit diverged again on GD 200 (λ rms 1.2×10^{-6}).
- **Circle**: Kn 0.5 shift -0.14 ± 0.03 (split), -0.10 ± 0.03 (two-sided), -0.12 ± 0.03 (per-query); free-molecular $+0.01 \pm 0.02$. S8 it 30: $\cos 2\theta$ $+0.08 \pm 0.03$ (Kn) and $+0.14 \pm 0.03$ *free-molecular*: a shape-dependent bias with no collision fit involved.
- **Elongated iterates**: the adjoint's per-seed sd is 3–13× the FD's in *both* regimes (heavy-tailed seeds); means within ~ 2 se. Split vs two-sided: indistinguishable.

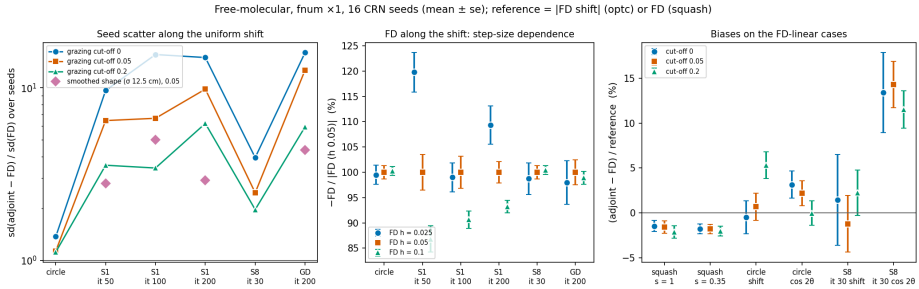
gapgate: Kn 0.5 — basis × horizon × run length

Same forward captured for its last 300 and 1000 steps, both fits paired; production length $T = 1000$ and a filled box $T = 2000$; 16 CRN seeds a point; `scratch/gapgate/analysis.py`.



- **Squash**: horizon $1000 - 300$ $+2.1 \pm 0.3\,\%$ at *both* T (adjoint memory, not the fill); `cubicv`−`quad` $+3.0 \pm 0.2\,\%$. Together they overshoot ($+1.9 \pm 0.65\,\%$): part of the cubic's gain stands in for the horizon. At 300 steps `cubicv` reads -0.7 / $-1.8\,\%$ (quad -3.7 / -5.0) $\Rightarrow$ **adopted**; stable on all optc iterates.
- **Circle shift**: $-14 \pm 3\,\%$ at $T = 1000$, $-1 \pm 4\,\%$ at $T = 2000$; the adjoint is the same at both, the FD moves 13 %; the whole run captured (from an empty box) changes nothing. The FD of the filling box sees the shape's effect on the fill, which the fits do not carry; not an error of the steady gradient.

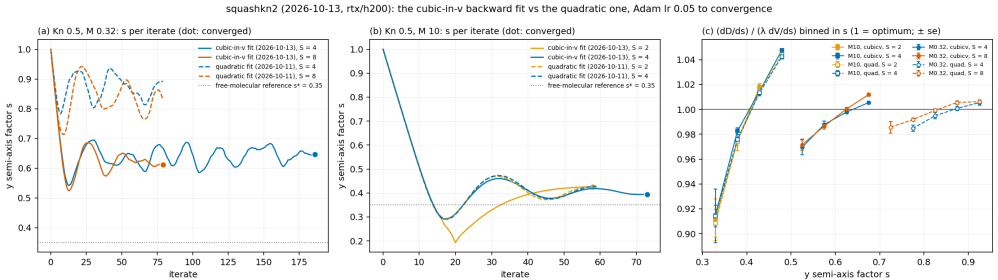
gapgate: free-molecular — grazing, smoothing, FD linearity



- **Variance = grazing hits on sub-cell features**: the scatter ratio falls with the cut-off (S1/GD: 10–16 / 6–12 / 3.5–6 at 0 / 0.05 / 0.2; circle ≈ 1 at any) and with smoothing the features (2.8–5).
- **Biases are not the cut-off**: exact − shipped, paired: squash $+0.10 \pm 0.23$ / $+0.01 \pm 0.07$ %, S8 $\cos 2\theta$ -0.01 ± 0.03. Still unlocated: squash ~ -1.5 %, S8 it 30 $\cos 2\theta$ $+0.14 \pm 0.03$ (FD linear).
- **Rough iterates**: FD shift at $h = 0.025$ / 0.05 / 0.1 moves 20–30 % (S1 50, 200); no derivative at that scale.

squashkn2: the Kn 0.5 optimum is ill-conditioned

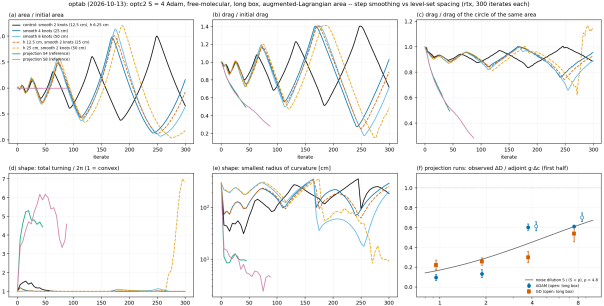
Adam, lr 0.05, to convergence (s of the last 5 iterates within 0.5 %); `scratch/optab/squash_analysis.py`.



- **Kn 0.5, M 0.32**: `cubicv` S4 / S8 converge to $s = 0.640 \pm 0.016$ / 0.623 ± 0.013 (last 30 iterates); the quadratic fit's runs sat at 0.83–0.86. The slope ratio $(\mathrm{d}D/\mathrm{d}s)/(\lambda\,\mathrm{d}V/\mathrm{d}s)$ is within 2 % of 1 over $s \in [0.5, 0.9]$ (0.27 per unit s at the crossing): the fit's $+3\,\%$ moved the optimum by 0.2, and the remaining $-1.2 \pm 0.6\,\%$ puts the true optimum $\approx 0.045 \pm 0.02$ *lower*, $s^* \approx 0.58$–0.62.
- **Kn 0.5, M 10**: both fits agree (crossing $s \approx 0.40$, slope ~ 0.9 per unit s: well conditioned); S4 0.400, S2 0.394 (stopped on a slow drift at 0.426).

optab: the augmented Lagrangian winds up

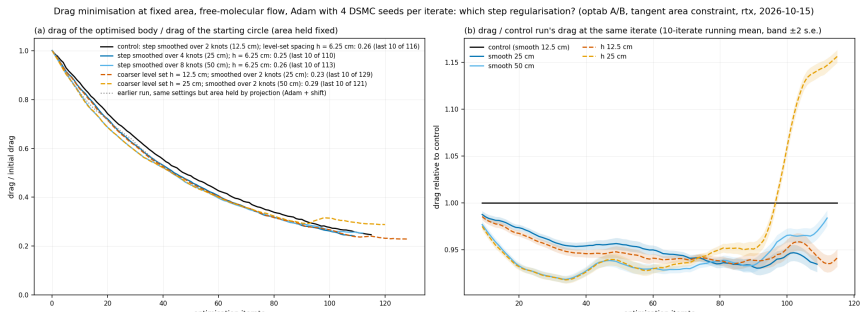
optc2 S4 Adam (free-molecular, long box), step smoothed 2 / 4 / 8 knots vs level set $h = 12.5$ / 25 cm (smooth 2 knots: the same physical widths); `scratch/optab/analysis.py`.



- **Area control unstable**: a fixed-size step overshoots whatever the pull, and $\nu \leftarrow \nu + \mu\,\Delta A$ integrates the overshoot — ν reached $\sim 400\times$ the price, the area swung 4–220 % of A_0, the shapes stayed convex blobs ($D/D_{\mathrm{circle}}(A)$ 0.86–0.93 vs 0.33 for the projection S8). Replaced by `--area-mode tangent`: exact area, the step projected onto $\nabla A^{\perp}$ (a multiplier per iterate), the shift only the $O(\mathrm{step}^2)$ remainder (0.5 mm vs 1–2 cm).
- **Smoothing** (first 40 iterates, before the swings): the control grows wiggles (turning 1.5, $r_{\min}$ 30–50 cm); every wider setting stays convex. At equal width the coarse level set and the wider smoothing are the same ($r_{\min} \approx 1$ m at 25 cm, ≈ 2 m at 50 cm): coarsening buys nothing (at 25 cm it kinked once the body shrank to ~ 1 m).
- **Apparent adjoint error** (panel f, projection runs): observed ΔD / predicted $= 0.1$–0.7, rising with S as $S/(S+\rho)$, $\rho = 4.8$: the step follows the noisy gradient, so $g \cdot \Delta c$ carries the noise's power. Noise, not bias.

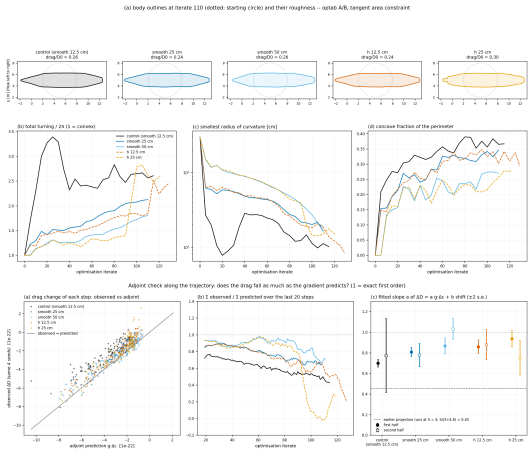
optab with the tangent area mode: drag at fixed area

The same five runs, `--area-mode tangent` (tags `circle_long_S4_adam_tan_ab_*`), first 110–129 of 300 iterates (pulled 2026-10-15); area held to 10^{-12}; `scratch/optab/analysis_tan.py`.

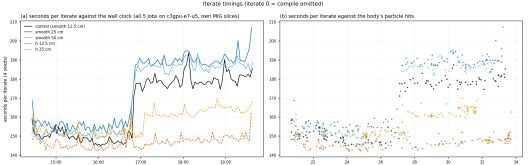


- Drag falls to **0.23–0.26 of the circle's** at equal area (the projection run's η was 0.33 at 300). All settings within $\sim$4 % of each other, except $h = 25$ cm, which kinked at its nose (iterate $\sim$95, $+15\,\%$).
- The wider settings lead the control by 5–8 % over iterates 10–80; the control closes the gap late.

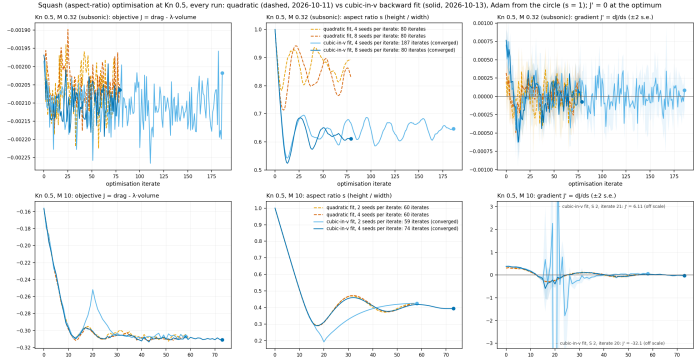
optab (tangent): shapes, adjoint check, timings



- **Shapes**: all become the same slender body, 15 m × 2.4 m (from a 6 m circle). Smoothing over 8 knots (50 cm) is the smoothest (turning 1.8, $r_{\min}$ 19 cm); the control the roughest (turning 2.6, $r_{\min}$ 10 cm, a blunt kinked tail).
- **Adjoint**: observed ΔD / predicted $g \cdot \Delta c = 0.70$ (control), 0.79 (25 cm), **0.88** (50 cm), correlation 0.90–0.97 — against 0.45 for the S4 projection runs. Adam's momentum takes most of the step's own noise out of $g \cdot \Delta c$. The 20-step ratio falls late in every run (0.4–0.7 by iterate 110): the gradient shrinks while the step stays 5 cm, so noise and the second-order term grow relative to the decrease — a step decay is due.
- **Timings**: 3 runs went from 150 to 180–190 s per iterate at the same minute (16:47) at different iterates (56–59), $h = 25$ cm +11 s at 17:02, $h = 12.5$ cm never. No change in hits; the step follows the clock, not the run: a neighbour on the shared node, not the code.



squashkn2: every run's history, and the M 10 cubic S2 spike



- **M 0.32**: `cubicv` converges at $s = 0.61$–0.65 (S4 oscillates ± 0.04, period ~ 25); the quadratic runs sat at 0.76–0.94 unconverged. J is flat there (noise-dominated): the ill-conditioning of the previous frames.
- **M 10**: all four end at $s \approx 0.39$–0.42. The `cubicv` S2 spike (iterate 20, $s = 0.193$: $\mathrm{d}D/\mathrm{d}s = -29.9 \pm 7.1$ vs the reference $+1.48$) is *only* `pathwise_v` (-31.4; tally $+1.47$, forward drag within 0.3 %), both seeds. $|$`pathwise_v`$|$ median 0.05 at $s \in [0.35, 0.45]$, 0.62 at $[0.25, 0.30]$, 2.0 below 0.25 (quadratic: 0.03): $\nabla_v \lambda$ of the cubic-in-v fit breaks down once the body is under ~ 2 DSMC cells thick. Adam's v jumped 0.002 $\rightarrow$ 1.04 and starved the later steps: the "converged" $s = 0.424$ is a step-starvation stop (reference $J' = +0.22$ there).

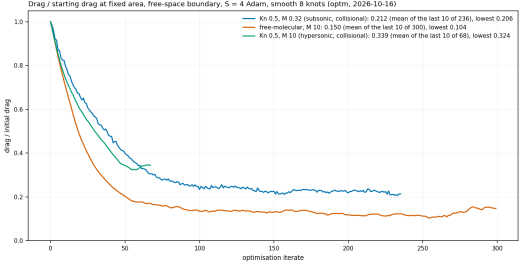
Free-space boundary: why, and its gates

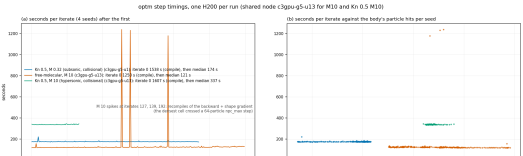
`forward.build(boundary="free")` (`--boundary free`): every box face injects the drifting Maxwellian's inward flux and removes what leaves; injection capacity $2\times$ the largest face's target; tags `_free`.

- **The channel holds a half-range stream**: free-molecular circle Cd 17.69 / 17.33 / 17.35 at blockage 0.3 / 0.15 / 0.075 against the half-range closed form 17.17; the full Maxwellian (free space) gives 12.22 at M 0.32.
- **The channel chokes at Kn 0.5, M 10**: a channel-wide region at $\sim 4.4\, n_\infty$ (the strong-shock ratio) grows upstream for $D/H \geq 0.3$; 30 m of upstream box only gave it room (x -14..0 m, 600 k slots overflowed). $D/H = 0.2$ starts (`scratch/optm10/m10_gate.py`). Free-molecular M 10's $1.9\, n_\infty$ ahead of the body is physical (specular back-reflection).
- **Free-space gates** (`scratch/freebox/`):
  - free-molecular Cd, full-Maxwellian start: $R = 1.5$: 12.245 ± 0.036 ($z = 0.8$); $R = 0.75$: 12.342 ± 0.056 ($z = 2.2$, 1 % high); counts steady, no inflow shortfall;
  - Kn 0.5 + M 10, $R = 3$ in `circle_long`: starts, $1.09\, n_\infty$ at the inflow face, ~ 2 ahead of the body, 117 k alive;
  - radius gradient, level-set circle, free-molecular, $f_{\mathrm{num}} \times 4$, 8 seeds (newborns from all four faces): $+2.2 \pm 2.7$ % vs D/R ($z = 0.8$), -0.9 ± 3.6 % vs FD 0.05 ($z = -0.3$), $+2.1 \pm 3.1$ % vs FD 0.1.

optm: drag at fixed area in free space, three regimes

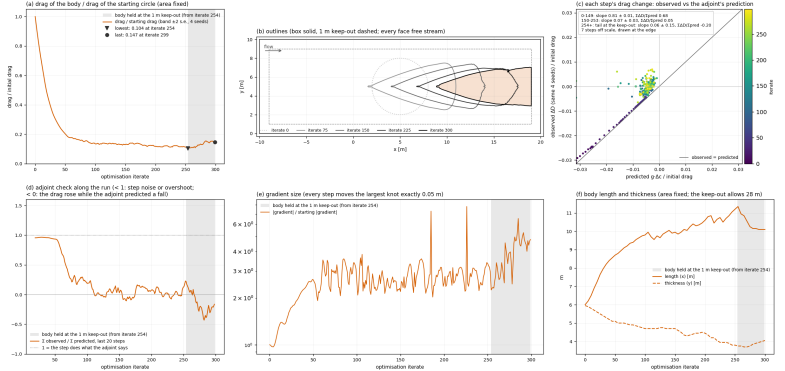
S4 Adam, smooth 8 knots, tangent area, `circle_long` (x $-10..20$, y 0..10 m), 1 m keep-out, `--boundary free`, one H200 each; pulled 2026-10-16 (Kn 0.5 236, free-mol. M 10 300 = done, Kn 0.5 + M 10 68 iterates); `scratch/optm/analysis_optm.py`.





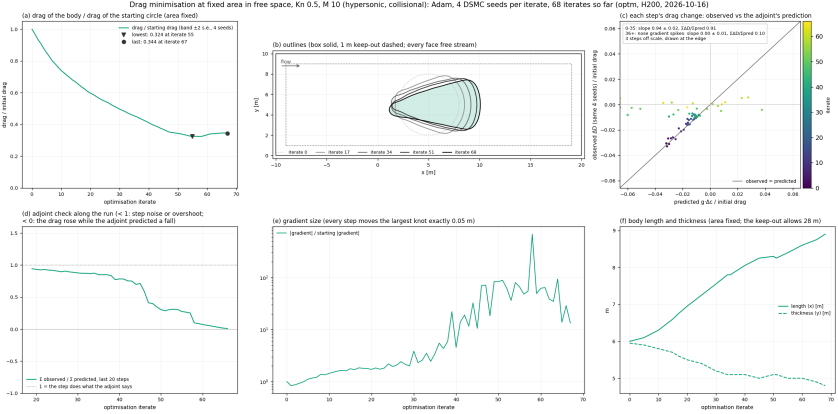
- **Kn 0.5, M 0.32**: 0.21 of the circle's drag, a slender 13.8 m $\times$ 2.9 m body; flat since $\sim$140. Adjoint slope (observed ΔD on $g \cdot \Delta c$) 0.81 ± 0.13 over 0–79, then within noise — the fixed 5 cm step at the noise floor: a step decay would pay here.
- **Timings** (H200, s per iterate): 174 / 121 / 337; iterate 0 1250–1610 (compile). Free-mol. M 10 iterates 127, 139, 192 took $\sim$1200 s: full recompiles of the backward and the shape gradient, the densest cell having crossed a 64-particle step of `npc_max` (a static argument; `NPC_BUCKET`). $\sim$55 min of 11.4 h. $+6$ s per iterate while the keep-out is active (floor / shift passes, not timed).

optm free-molecular M 10: the box, not the step



- The body becomes a wedge with a flat base. At M 10 nothing hits the base, so area there costs no drag: the area constraint moves area into the shadow and the body slides downstream (nose x 1.7 → 8.9 m, length ∼10 m throughout).
- Lowest drag **0.104** at iterate 254 — the iterate the base reached the $x = 19$ m keep-out. From there the floor clips ∼2300 knots per step and the area restore thickens the wedge (3.75 → 4.05 m): drag 0.15 at 300 while the adjoint still predicted a fall every step.
- Before the keep-out (150–253): slope 0.07 ± 0.03 at an objective s.e. of 0.04 %: not noise — overshoot of the fixed 5 cm step or a gradient error at the sharp nose ($r_{\min} \approx 5$ cm). Resuming from `coef_250` at half the step separates them.

optm Kn 0.5 + M 10: gradient spikes at the nose



- Iterates 0–35: slope 0.94 ± 0.02, drag 0.50. From $\sim$36 $|g|$ grows to 70–680$\times$ its start; slope ≈ 0; drag stalls at 0.324 (iterate 55) and creeps to 0.344.
- The spikes sit at the nose (x $\approx$ 1.5, y $\approx$ 4.2 m, just inside the surface): Adam's v there $\sim$790$\times$ the surface median, so the nose's steps starve while the rest move. Same signature as the squash `cubicv` spike; the per-step gradient is not saved, so the term is not yet identified.

Validation status of the current estimator

case	result	regime
parameter gradients (inlet, Couette walls, VSS kernel), virtual 4×8	$\lvert z\rvert \le 1.7$; $\le 1\,\%$ bias	collisional
Rankine–Hugoniot $\mathrm{d}(\rho_2/\rho_1)/\mathrm{d}M_1$	$z = 0.3$ (coarse)	collisional
level-set circle radius; sphere radius	$z = -1.4$; $z = -1.71$	free-mol.
$\cos 2\theta$ coarse, state-based fits	paired $+0.007 \pm 0.071$	free-mol.
diffuse plate (silhouette + hybrid normal)	$z = +0.19$ ($\lvert z\rvert \le 1.6$ at 5 steps)	free-mol.
squash specular, $s = 0.35/0.6/1$	$-1.7/-2.5/-1.5\,\%$ (± 1.5); $\le 1\,\%$ is horizon	free-mol.
squash $s = 1$, diffuse	$-1.4 \pm 3.3\,\%$	free-mol.
cut-volume machinery	$z = -1.2$ (all cells), $+0.8$ (cut cells)	Kn 0.5
squash Kn 0.5, $f_{\mathrm{num}} \times 2$ (vs Richardson FD) / $\times 4$ (12 seeds)	-7.5 / $-1.9 \pm 2.3\,\%$	superseded
squash $s = 1$, Kn 0.5, $f_{\mathrm{num}} \times 1$, $T = 1000$ / 2000: `quad` / `cubicv`	-3.7, -5.0 / -0.7, $-1.8\,\%$ (± 0.8)	Kn 0.5
optc circle shift, Kn 0.5 $T = 1000$ / 2000; free-mol.	-14 ± 3 / $-1 \pm 4\,\%$; $+1 \pm 2\,\%$	both
free-space boundary: Cd $R = 1.5$ / 0.75; radius gradient (8 seeds)	$z = 0.8$ / 2.2; $+2.2 \pm 2.7\,\%$ (FD -0.9 ± 3.6)	free-mol.

Free-molecular shape gradients agree with FD to $\sim$2 % (the horizon $\le 1\,\%$ of it). Kn 0.5 at production density with the shipped cubic-in-v fit: $-1.2 \pm 0.6\,\%$ (two run lengths pooled). Parameter gradients on collisional flows: $\le 1\,\%$, either fit.

Open items

1. **Kn 0.5**: `cubicv` at 300 steps $-1.2 \pm 0.6\,\%$; with the whole run $+1.9 \pm 0.65\,\%$ — the fit's and the horizon's corrections overlap; a longer-horizon default would need the fit revisited.
2. **Free-molecular remainder**: squash $\sim -1.5\,\%$ and S8 it 30 $\cos 2\theta$ $+0.14 \pm 0.03$; not grazing, horizon or FD.
3. **Shape-gradient noise**: grazing hits on sub-cell features the few-seed runs grow; prevent the features rather than raise the cut-off (0.2 biases the circle $+5\,\%$). `optab`: a step smoothing of 4–8 knots (25–50 cm) keeps the early iterates convex; a coarser level set adds nothing. Per-step signal fraction $S/(S+4.8)$ in projection mode; 0.7–0.9 with the tangent mode's Adam runs. Choice: smooth 8 knots, $h = 6.25$ cm.
4. **Area**: `--area-mode tangent` (projected gradient: $g - \nu \nabla A$, $\nu = g \cdot \nabla A / |\nabla A|^2$, the step projected too) replaces the wound-up augmented Lagrangian; the restoring shift is 0.5 mm, against 1–2 cm per Adam iterate in the old projection mode. `optab` (tags `_tan_ab_*`): area to 10^{-12}, drag 0.23–0.26 of the circle's.
5. **Re-runs**: free-molecular squash and `optc` can be re-run now (the old runs used `along_pool`, $-9\,\%$ at $s = 0.35$, and `optc` the future class); squashkn after item 1.
6. **`optc` bodies** need a body-to-face clearance constraint before their drag reductions are quoted: the `optc2` and `optckn` re-runs use a 3× longer box centred on the body (`circle_long`, 15 m each side; warm-up 2800 steps) and keep the body 1 m from every face (`optimize --keepout`).

Open items (2026-10-16)

7. **Free space** (`--boundary free`, 2026-10-15): gated; used by `optm`. Retrofit to `optc2`, `optckn`, `optab` and the squash groups (new tags) still to do. The 10 m box leaves 2 m between the faces and a 6 m body.
8. **Gradient blow-ups**: squash M 10 `cubicv` below ∼2 cells thick (`pathwise_v`); `optm` Kn 0.5 + M 10 at the nose from iterate 36. Locate the term (save the per-knot gradient), then a guard; Adam's v starves steps after a spike.
9. **Box-bound optima**: free-molecular M 10 moves area into the drag-free base until the keep-out stops it (`optm` 254). A bigger box or a base constraint before quoting the wedge's drag.
10. **Step**: fixed 5 cm per iterate; slope falls to 0.05–0.2 late in every run. Step decay not built.
11. **Recompiles**: `npc_max` bucket of 64 crossed → ∼1200 s on H200 (3× in 300 iterates).
12. Translations of flat walls remain unsupported as controls.

Summaries: `scratch/bias/summarise.py`, `horizon_summary.py`; cluster: `examples/mesh/cluster/README.md` (`biasgate`).